



# The Binomial Theorem

## Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 Find the term independent of  $x$  in the expansion of  $(2x^2 - \frac{3}{x})^9$ . 2

$$\begin{aligned}
 & (2x^2 - 3x^{-1})^9 \\
 T_k &= \binom{9}{k} (2x^2)^{9-k} (-3x^{-1})^k, \quad k=0,1,2,\dots,9 \\
 &= \binom{9}{k} \cdot 2^{9-k} \cdot x^{18-2k} \cdot (-3)^k \cdot x^{-k} \\
 &= \binom{9}{k} \cdot 2^{9-k} \cdot (-3)^k \cdot x^{18-3k} \\
 \text{Indep. of } x: & 18-3k=0 \Rightarrow k=6 \\
 T_6 &= \binom{9}{6} \cdot 2^3 \cdot (-3)^6 \cdot x^0 = 989888 \quad [2]
 \end{aligned}$$

- 2 Find the term independent of  $x$  in the expansion of  $(3x - \frac{1}{2x^2})^9$ . 3

$$\begin{aligned}
 \left(3x - \frac{1}{2x^2}\right)^9 &= \sum_{k=0}^9 \binom{9}{k} (3x)^{9-k} \left(-\frac{1}{2x^2}\right)^k \\
 &= \sum_{k=0}^9 \binom{9}{k} 3^{9-k} \left(-\frac{1}{2}\right)^k x^{9-k} \times x^{-2k} \\
 &= \sum_{k=0}^9 \binom{9}{k} 3^{9-k} \left(-\frac{1}{2}\right)^k x^{9-3k}
 \end{aligned}$$

The term independent of  $x$  is when  $k=3$ .

$$\therefore \binom{9}{3} 3^6 \left(-\frac{1}{2}\right)^3 = -7654.5 \text{ or } -\frac{15309}{2}$$

- 3 Find the coefficient of  $x$  in the expansion of  $(2x - \frac{1}{2x})^9$ . 2

$$Ax = {}^9C_r (2x)^{9-r} \left(-\frac{1}{2x}\right)^r = (-1)^r \times {}^9C_r \times 2^{9-r} 2^{-r} \times x^{9-r} x^{-r} = (-1)^r {}^9C_r 2^{9-2r} x^{9-2r}$$

Hence if we solve the powers for  $x$ , we get  $9-2r=1$ , and  $r$  is clearly 4.  
we will get  ${}^9C_4 2x = 252x$  and the coefficient of  $x$  is 252.

- 4 By considering  $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$ , show that 2

$$n2^{n-1} = \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \dots + n\binom{n}{n}$$

Differentiate both sides wrt  $x$ :  $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$

$$\therefore n(1+x)^{n-1} = {}^nC_1 + 2{}^nC_2x + \dots + n{}^nC_nx^{n-1}$$

Substitute  $x=1$ :  $n(1+1)^{n-1} = {}^nC_1 + 2{}^nC_2 + \dots + n{}^nC_n$

$$\therefore n2^{n-1} = {}^nC_1 + 2{}^nC_2 + \dots + n{}^nC_n$$

5 By using the fact that  $(1+x)^{11} = (1+x)^3(1+x)^8$ . Show that

3

$$\binom{11}{5} = \binom{8}{5} + \binom{3}{1}\binom{8}{4} + \binom{3}{2}\binom{8}{3} + \binom{8}{2}$$

$$(1+x)^{11} = (1+x)^3(1+x)^8$$

General Term of  $(1+x)^{11}$

$$T_k = {}^{11}C_k x^{11-k}$$

$\therefore {}^{11}C_5$  is the coefficient of  $x^5$

$$(1+x)^3 = {}^3C_0 + {}^3C_1x + {}^3C_2x^2 + {}^3C_3x^3$$

$$(1+x)^8 = {}^8C_0 + {}^8C_1x + {}^8C_2x^2 + {}^8C_3x^3 + {}^8C_4x^4 + {}^8C_5x^5 + \dots$$

$\therefore$  coefficient of  $x^5$  in  $(1+x)^3(1+x)^8$

$$= {}^3C_0 \times {}^8C_5 + {}^3C_1 \times {}^8C_4 + {}^3C_2 \times {}^8C_3 + {}^3C_3 \times {}^8C_2$$

$$= {}^8C_5 + {}^3C_1 \times {}^8C_4 + {}^3C_2 \times {}^8C_3 + {}^8C_2$$

Equating coefficients

$$\binom{11}{5} = \binom{8}{5} + \binom{3}{1}\binom{8}{4} + \binom{3}{2}\binom{8}{3} + \binom{8}{2}$$

6 The coefficient of  $x^3$  in the expansion of  $(a+x)^6$  is 1280, what is the value of  $a$ ?

1

$$T_{k+1} = {}^6C_3 a^3 x^3$$

$$1280 = {}^6C_3 a^3$$

$$\sqrt[3]{\frac{1280}{{}^6C_3}} = a$$

$$a = 4$$

7 Find the term independent of  $x$  in the expansion  $\left(x^2 - \frac{1}{x^2}\right)^{10}$ .

2

$$T_{k+1} = {}^{10}C_k (x^2)^{10-k} \left(-\frac{1}{x^2}\right)^k$$

$$= {}^{10}C_k x^{20-2k} (-1)^k x^{-2k}$$

$$= {}^{10}C_k x^{20-4k} (-1)^k$$

$$\text{Term indep. of } x \Rightarrow 20-4k = 0$$

$$4k = 20$$

$$k = 5$$

$$\therefore T_6 = {}^{10}C_5 (-1)^5 = -{}^{10}C_5 = -252$$

8 Find the coefficient of  $x^4$  in the expansion  $\left(x + \frac{3}{x}\right)^8$ .

2

$$T_{k+1} = {}^8C_k (x)^{8-k} \left(\frac{3}{x}\right)^k = {}^8C_k x^{8-k} (3^k x^{-k}) = {}^8C_k (3)^k x^{8-2k}$$

$$x^4 \Rightarrow 8-2k = 4 \quad k = 2$$

$$\text{the term is } {}^8C_2 \times (3)^2 = 252$$

9 What is the coefficient of the fourth term in the expansion of  $(3x-4)^6$ ?

2

$$(3x-4)^6 = {}^6C_0 (3x)^6 (-4)^0 + \dots + {}^6C_3 (3x)^3 (-4)^3$$

$$\text{coefficient} = {}^6C_3 \times 3^3 \times (-4)^3 = -34560$$

- 10 The first three terms of the binomial expansion of  $(1+ax)^n$  are  $1 - 4x + \frac{20}{3}x^2$ . Find the values of  $a$  and  $n$ . 3

$$\begin{aligned}(1+ax)^n &= \binom{n}{0}(ax)^0 + \binom{n}{1}(ax)^1 + \binom{n}{2}(ax)^2 + \dots \\ &= 1 + anx + \frac{n!}{(n-2)!2!}a^2x^2 + \dots\end{aligned}$$

$$\begin{aligned}-4x &= anx \\ -4 &= an \\ a &= \frac{-4}{n}\end{aligned}$$

$$\begin{aligned}\frac{20}{3}x^2 &= \frac{n!}{(n-2)!2!}a^2x^2 \\ \frac{20}{3} &= \frac{n(n-1)(n-2)!}{(n-2)!2}a^2 \\ \frac{20}{3} &= \frac{n(n-1)}{2}a^2 \quad a = \frac{-4}{n} \\ \frac{20}{3} &= \frac{n(n-1)}{2} \left(\frac{16}{n^2}\right) \\ \frac{20}{3} &= \frac{(n-1)(8)}{n} \\ \frac{5}{3} &= \frac{(n-1)2}{n} \\ 5n &= 6(n-1) \\ 5n &= 6n-6 \quad \boxed{n=6}\end{aligned}$$

$$\begin{aligned}a &= \frac{-4}{n} \\ a &= \frac{-4}{6} \\ \boxed{a = -\frac{2}{3}}\end{aligned}$$

- 11 What is the coefficient of  $x^9$  in  $(x+5)^{16}$ ? 1

$$\begin{aligned}{}^{16}C_9 (x)^9 (5)^{16-9} \\ {}^{16}C_9 5^7 x^9\end{aligned}$$

- 12 Evaluate  $\binom{n-1}{k-1} + \binom{n-1}{k}$  1

A.  $\frac{n!}{(n-k)!}$

B.  $\frac{n!}{k!(n-k)!}$

C.  $\frac{n!}{(k-1)!(n-(k-1))!}$

D.  $\frac{n!}{(n-(k-1))!}$

$$\begin{aligned}&= \frac{(n-1)!}{(n-1-(k-1))!(k-1)!} + \frac{(n-1)!}{((n-1)-k)!k!} \\ &= \frac{(n-1)!}{(n-1-k+1)(n-1-k)!(k-1)!} + \frac{(n-1)!}{(n-1-k)!k(k-1)!} \\ &= \frac{k(n-1)! + (n-k)(n-1)!}{(n-k)!k!} \\ &= \frac{k(n-1)! + n(n-1)! - k(n-1)!}{(n-k)!k!} = \frac{n!}{(n-k)!k!}\end{aligned}$$

- 13 Expand  $(1+x)^{2n} + (1-x)^{2n}$  and hence, evaluate 2

$$\begin{aligned}1 + \binom{100}{2} + \binom{100}{4} + \dots + \binom{100}{100} \\ (1+x)^{2n} + (1-x)^{2n} &= \binom{2n}{0}x^0 + \binom{2n}{1}x^1 + \binom{2n}{2}x^2 + \dots + \binom{2n}{2n}x^{2n} + \binom{2n}{0}x^0 - \binom{2n}{1}x + \binom{2n}{2}x^2 - \dots + \binom{2n}{2n}x^{2n} \\ &= 1 + \binom{2n}{2}x^2 + \binom{2n}{4}x^4 + \dots + \binom{2n}{2n}x^{2n} + 1 - \binom{2n}{1}x + \binom{2n}{2}x^2 - \binom{2n}{3}x^3 + \dots + \binom{2n}{2n}x^{2n} \\ &= 2[1 + \binom{2n}{2}x^2 + \binom{2n}{4}x^4 + \dots + \binom{2n}{2n}x^{2n}] \\ \text{So, } \frac{1}{2}[(1+x)^{2n} + (1-x)^{2n}] &= 1 + \binom{2n}{2}x^2 + \binom{2n}{4}x^4 + \dots + \binom{2n}{2n}x^{2n} \\ \text{when } n=50 \text{ and } x=1, \text{ then } \text{RHS} &= [1 + \binom{100}{2} + \binom{100}{4} + \dots + \binom{100}{100}] \\ \text{So, LHS} &= \frac{1}{2}[(1+1)^{100} + (1-1)^{100}] = \frac{1}{2} \cdot 2^{100} = 2^{99}\end{aligned}$$

14 Use the identity  $(1+x)^m(1+x)^n = (1+x)^{m+n}$  to show that

2

$$\binom{m+n}{4} = \binom{n}{4} + \binom{n}{3}\binom{m}{1} + \binom{n}{2}\binom{m}{2} + \binom{n}{1}\binom{m}{3} + \binom{m}{4}$$

Consider the identity  $(1+x)^m(1+x)^n = (1+x)^{m+n}$ .

The term  $x^4$  of  $(1+x)^{m+n}$  is:  $\binom{m+n}{4}x^4$  (1)

The term  $x^4$  of  $(1+x)^m(1+x)^n$  is:

$$\begin{aligned} & \binom{m}{4}x^4 \times 1 + \binom{m}{3}x^3 \times \binom{n}{1}x + \binom{m}{2}x^2 \times \binom{n}{2}x^2 + \binom{m}{1}x \times \binom{n}{3}x^3 + 1 \times \binom{n}{4}x^4 \\ &= \left[ \binom{m}{4} + \binom{m}{3}\binom{n}{1} + \binom{m}{2}\binom{n}{2} + \binom{m}{1}\binom{n}{3} + \binom{n}{4} \right] x^4 \quad (2) \end{aligned}$$

Compare the coefficients of  $x^4$  from (1) and (2):

$$\binom{m+n}{4} = \binom{m}{4} + \binom{m}{3}\binom{n}{1} + \binom{m}{2}\binom{n}{2} + \binom{m}{1}\binom{n}{3} + \binom{n}{4}$$

15 Write the expansion of  $(2x-y)^5$ .

2

$$\begin{aligned} (2x-y)^5 &= (2x)^5 - {}^5C_1(2x)^4y + {}^5C_2(2x)^3y^2 - {}^5C_3(2x)^2y^3 + {}^5C_4(2x)y^4 - y^5 \\ &= 32x^5 - 80x^4y + 80x^3y^2 - 40x^2y^3 + 10xy^4 - y^5 \end{aligned}$$



# The Binomial Theorem

## Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 Find the constant term in the expansion:  $(4x^4 - \frac{2}{x^2})^9$  3

$$\begin{aligned} \left(4x^4 - \frac{2}{x^2}\right)^9 &= {}^9C_0 (4x^4)^9 + {}^9C_1 (4x^4)^8 \left(-\frac{2}{x^2}\right)^1 \\ &\quad + {}^9C_2 (4x^4)^7 \left(-\frac{2}{x^2}\right)^2 + \dots \\ &\quad + {}^9C_6 (4x^4)^3 \left(-\frac{2}{x^2}\right)^6 \\ &\quad + {}^9C_7 (4x^4)^2 \left(-\frac{2}{x^2}\right)^7 + {}^9C_8 (4x^4)^1 \left(-\frac{2}{x^2}\right)^8 + {}^9C_9 (4x^4)^0 \left(-\frac{2}{x^2}\right)^9 \end{aligned}$$

$24 \times 64 \times 64 \quad \boxed{= 344064}$

- 2 a. Find the general term of  $(x + \frac{1}{x})^8$ . 1

$${}^8C_r (x)^{8-r} \left(\frac{1}{x}\right)^r = {}^8C_r (x)^{8-2r}$$

- b. Find the coefficient of  $x^2$  in  $(x + \frac{1}{x})^8$ . 2

For the term with  $x^2$   $8 - 2r = 2 \quad r = 3$

The term is:  ${}^8C_3 (x)^{8-3} \left(\frac{1}{x}\right)^3 = 56x^2$

The coefficient of  $x^2$  is 56

- 3 The coefficients of the  $x^{17}$  and the  $x^{27}$  terms are equal in the expansion of  $(px^3 - \frac{q}{x^2})^9$ . Prove 3

that  $p - 6q = 0$ , where  $p$  and  $q$  are positive integers.

$$\left(px^3 - \frac{q}{x^2}\right)^9 = \sum_{r=0}^9 {}^9C_r (px^3)^{9-r} (-q/x^2)^r = \sum_{r=0}^9 {}^9C_r (-1)^r p^{9-r} q^r x^{27-5r}$$

For the  $x^{17}$  term:

$$27 - 5r = 17$$

$$r = 2$$

$$c_2 = {}^9C_2 p^7 q^2$$

For the  $x^{27}$  term:

$$27 - 5r = 27$$

$$r = 0$$

$$c_0 = {}^9C_0 p^9$$

Given  $c_0 = c_2$ :  $(1)p^9 = (36)p^7 q^2$

$$p^9 - 36p^7 q^2 = 0$$

$$p^7(p^2 - 36q^2) = 0$$

$$p^7(p - 6q)(p + 6q) = 0$$

$$p = 0, p - 6q = 0 \text{ or } p + 6q = 0$$

But given  $p > 0$  and  $q > 0$ . Hence  $p - 6q = 0$  only.

- 4 a. Write down the expansion of  $(1 - x)^6$ . 1

$$\begin{aligned} (1-x)^6 &= (1+(-x))^6 \\ &= {}^6C_0 (-x)^0 + {}^6C_1 (-x)^1 + {}^6C_2 (-x)^2 + {}^6C_3 (-x)^3 + {}^6C_4 (-x)^4 + {}^6C_5 (-x)^5 + {}^6C_6 (-x)^6 \\ &= 1 - 6x + 15x^2 - 20x^3 + 15x^4 - 6x^5 + x^6 \end{aligned}$$

- b. Hence, find the term in  $x^2$  in the expansion  $(2x - 1)^2(1 - x)^6$ .

$$(2x - 1)^2 = 4x^2 - 4x + 1$$

$\therefore$  The term in  $x^2$

$$= 4x^2 \times 1 - 4x \times (-6x) + 1 \times 15x^2 = 43x^2$$

- 5 A box contains 16 red, 10 blue and 12 yellow balls. Use the pigeonhole principle to find the minimum number of balls to be drawn from the box to ensure 9 balls of same colour. 2

$$\begin{aligned} \text{No. of colours (pigeonholes)} &= 3 \\ \therefore \text{Min. no. of balls} &= 3 \times 8 + 1 = 25 \\ (\text{eg: } 8 \text{ red} + 8 \text{ blue} + 8 \text{ yellow} + 1 (\text{r or b or y})) \end{aligned}$$

- 6 a. Use the binomial theorem to find an expression for the constant term in the expansion 3

of  $(2x^3 - \frac{1}{x})^{12}$ .

$$(2x^3 - \frac{1}{x})^{12} = \sum_{r=0}^{12} \binom{12}{r} (2x^3)^r (-\frac{1}{x})^{12-r}$$

for constant  $x^{3r} \cdot x^{-(12-r)} = x^0$   
 $\therefore 3r + (r-12) = 0$   
 $4r = 12$   
 $r = 3$

$\therefore$  constant term is  $\binom{12}{3} 2^3 (-1)^9 = -1760$

- b. For what values of  $n$  does  $(2x^3 - \frac{1}{x})^{12}$  have a non-zero constant term? 1

$$\begin{aligned} x^{3r} \cdot (x^{-1})^{n-r} &= x^0 \\ 3r - n + r &= 0 \\ r &= \frac{n}{3} \end{aligned}$$

$\therefore (2x^3 - \frac{1}{x})^n$  will have a non-zero constant when  $n$  is divisible by 3

- 7 What is the coefficient of  $x^7$  in the expansion of  $(x + \frac{4}{x})^{13}$ ? 1

$$\begin{aligned} \binom{13}{r} x^{13-r} \left(\frac{4}{x}\right)^r & \quad 13-r-r=7 \quad \binom{13}{3} x^{10} \left(\frac{4}{x}\right)^3 = 18304x^7 \\ 6 &= 2r \\ r &= 3 \end{aligned}$$

- 8 Find the value of the constant  $a$  such that  $8750x^4$  is a term in the binomial expansion of 2

$(a + 2x)^7$ .

$$\begin{aligned} {}^7C_4 (2x)^4 a^3 &= 560a^3x^4 \quad [1 \text{ mark}] \\ a^3 &= \frac{8750}{560} = \frac{125}{8} \quad [1 \text{ mark}] \\ a &= \frac{5}{2} \quad [1 \text{ mark}] \end{aligned}$$

- 9 a. Write down the expansion of  $(1 + x)^3$ . 1

$$(1+x)^3 = 1 + 3x + 3x^2 + x^3$$

- b. Hence, find the coefficient of  $x^4$  from  $(2x - 1)^2(1 + x)^3$ . 2

$$\begin{aligned} (2x-1)^2(1+x)^3 &= (4x^2 - 4x + 1)(1+x)^3 \\ &= (4x^2 - 4x + 1)(1 + 3x + 3x^2 + x^3) \\ \text{coefficient of } x^4 &= 4 \times 3 - 4 \times 1 = 12 - 4 = 8 \end{aligned}$$

- 10 Find the coefficient of  $x^8$  in the expansion of  $\left(\frac{x}{4} + \frac{4}{x}\right)^{14}$ . 2

$$T_{k+1} = {}^{14}C_k \left(\frac{x}{4}\right)^{14-k} \left(\frac{4}{x}\right)^k$$

$$T_{k+1} = {}^{14}C_k \frac{(x)^k}{(4)^{14-k}} (x)^{14-2k}$$

$$T_4 = {}^{14}C_3 \frac{(x)^3}{(4)^{14-3}} (x)^{14-2 \times 3}$$

$$= \frac{{}^{14}C_3}{(4)^8} = \frac{91}{16384}$$

$14-2k=8$   
 $2k=14-8$   
 $2k=6$   
 $k=3$

- 11 a. Write out the first few terms in the expansion of  $(1-x)^n$ . 1

$$\binom{n}{0} - \binom{n}{1}x + \binom{n}{2}x^2 - \binom{n}{3}x^3 + \dots$$

- b. By an appropriate substitution, prove that: 2

$$\binom{n}{0} - 3 \times \binom{n}{1} + 9 \times \binom{n}{2} - 27 \times \binom{n}{3} + \dots = (-2)^n$$

Recognise that the substitution  $x = 3$  is needed.

$$\binom{n}{0} - \binom{n}{1} \times 3 + \binom{n}{2} \times (3)^2 - \binom{n}{3} \times (3)^3 + \dots = (1-3)^n.$$

$$\binom{n}{0} - 3 \times \binom{n}{1} + 9 \times \binom{n}{2} - 27 \times \binom{n}{3} + \dots = (-2)^n.$$

- 12 Differentiate the identity  $(1-x)^n = \binom{n}{0} - \binom{n}{1}x + \binom{n}{2}x^2 - \binom{n}{3}x^3 + \dots + \binom{n}{n}(-x)^n$ , then make a substitution to prove that: 3

$$-1 \times \binom{n}{1} + 6 \times \binom{n}{2} - 27 \times \binom{n}{3} - \dots + n \times 3^{n-1} \binom{n}{n} = -n(-2)^{n-1}$$

Differentiate the binomial expansion of  $(1-x)^n$  gives:

$$-nC_1 + 2nC_2x - 3nC_3x^2 - \dots + n(nC_n)x^{n-1} = -n(1-x)^{n-1}$$

Make the substitution of  $x = 3$ .